Group 1 Database-Specific Supplement - Apache CouchDB

Use with the DATS 6102 Final Group Project Class Overview | Document database; JSON documents over HTTP/REST

Important: This supplement contains only the requirements that are different for your assigned database. All common deadlines, grading, proposal/check-in requirements, final report requirements, presentation timing, demo grading, peer review, and AI policy are in the class-wide Final Group Project Overview and are not repeated here.

Group Members

Student	Assigned Database
Damian Mendoza	Apache CouchDB
Ingri Quevedo	Apache CouchDB
Stephanie Armstrong	Apache CouchDB
Hassan Saleh H Alshamrani	Apache CouchDB

Your Group Challenge

CouchDB emphasizes JSON documents, HTTP-based access, revision-based concurrency, and replication. Your group should show how this document-oriented approach differs from normalized relational design in MySQL.

Four-member group fairness: This group has four students. I selected CouchDB because it provides a manageable installation and implementation path. The academic requirements and grading rubric remain the same, but responsibilities may be combined. Peer review is based on average ratings received, so this group is not penalized for having fewer teammates.

Database-Specific Topics You Must Learn and Explain

- Document model, databases/documents, document IDs, and JSON structure.
- HTTP/REST interface and how applications interact with CouchDB.
- MVCC/concurrency behavior and document revision concepts.
- Replication, availability, and distributed/offline-first design concepts.
- Querying and indexing using Mango selectors/indexes and/or views as appropriate.
- Security/authentication basics, deployment options, licensing, strengths, limitations, and common use cases.

Database-Specific Technical Implementation

- Create a CouchDB database and load an appropriate JSON dataset.
- Design document structures that fit the scenario; justify embedding versus references where applicable.
- Demonstrate create, read, update, and delete operations.
- Create at least one useful index or query structure and explain why it is needed.
- Demonstrate at least four meaningful queries/operations, including a nontrivial multi-field retrieval.
- Demonstrate one CouchDB-specific capability such as replication/revision behavior or another feature I approve.

Required MySQL / Relational Comparison for This Database

Use your actual project scenario and data model; do not submit only a generic SQL-versus-NoSQL comparison.

- Map the CouchDB document design to a possible normalized MySQL design.
- Explain how document IDs and references compare with primary and foreign keys.
- Compare JSON/document retrieval with SQL joins and normalized tables.
- Compare concurrency/revision behavior and transaction scope with MySQL.

- Conclude whether the project scenario is better suited to CouchDB or MySQL and why.

NoSQL Comparison

Compare your assigned database with at least one NoSQL database covered in class (MongoDB or Neo4j). Focus on data model, querying, relationships, consistency/transactions, scaling, and use-case fit. Explain when each technology would be preferable.

Database-Specific Demo Checklist

- Show the database and loaded JSON documents.
- Show CRUD operations and at least four meaningful queries/operations.
- Show the index/query structure used by the project.
- Show one CouchDB-specific capability such as revisions or replication.

Suggested Division of Responsibilities

These are suggested primary responsibilities, not isolated assignments. Every member must understand the overall project and be prepared for individual questions.

- Architecture/research + technical documentation
- Installation/configuration + data loading
- Data modeling + queries/testing
- MySQL comparison + integration/presentation coordination

For all common requirements and grading details, return to the DATS 6102 Final Group Project Class Overview.